P2 Chapter 11: Integration

Using Further Substitution

SKILL #5: Integration by Substitution

For some integrations involving a complicated expression, we can make a substitution to turn it into an equivalent integration that is simpler. We wouldn't be able to use 'reverse chain rule' on the following:

Q Use the substitution $u = 2x + 5$ to find $\int x\sqrt{2x+5}\ dx$

The aim is to completely remove any reference to x, and replace it with u. We'll have to work out x and dx so that we can replace them.

STEP 1: Using substitution, work out x and dx (or variant)

$$\frac{du}{dx} = 2 \quad \rightarrow \quad dx = \frac{1}{2}du$$

$$x = \frac{u-5}{2}$$

Fro Tip: Be careful about ensuring you reciprocate when rearranging.

STEP 2: Substitute these into expression.

$$\int x\sqrt{2x+5}\ dx = \int \frac{u-5}{2}\sqrt{u}\frac{1}{2}du = \frac{1}{4}\int \sqrt{u}(u-5)\ du$$

$$= \frac{1}{4}\int u^{\frac{3}{2}} - 5u^{\frac{1}{2}}$$

Fro Tip: If you have a constant factor, factor it out of the integral.

STEP 3: Integrate simplified expression.

$$= \frac{1}{4}\left(\frac{2}{5}u^{\frac{5}{2}} - \frac{10}{3}u^{\frac{3}{2}}\right) + C$$

STEP 4: Write answer in terms of x.

$$= \frac{(2x+5)^{\frac{5}{2}}}{10} - \frac{5(2x+5)^{\frac{3}{2}}}{6} + C$$

How can we tell what substitution to use?

There's no hard and fast rule, but it's often helpful to replace to replace expressions inside roots, powers or the denominator of a fraction.

Good judgement requires the experience of trying!

Sensible substitution:

$$\int \cos x \sqrt{1+\sin x}\, dx \qquad \boldsymbol{u} = \boxed{?}$$

$$\int 6x\, e^{x^2}\, dx \qquad \boldsymbol{u} = \boxed{?}$$

$$\int \frac{xe^x}{1+x}\, dx \qquad \boldsymbol{u} = \boxed{?}$$

$$\int e^{\frac{1-x}{1+x}}\, dx \qquad \boldsymbol{u} = \boxed{?}$$

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There's no hard and fast rule, but it's often helpful to replace to replace expressions inside roots, powers or the denominator of a fraction.

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	Sensible substitution:	
$\int \cos x \sqrt{1+\sin x}\, dx$	$\boldsymbol{u = 1 + \sin x}$	
$\int 6x\, e^{x^2} dx$	$\boldsymbol{u = x^2}$	But this can be integrated by inspection.
$\int \frac{xe^x}{1+x} dx$	$\boldsymbol{u = 1 + x}$	
$\int e^{\frac{1-x}{1+x}} dx$	$\boldsymbol{u = \frac{1-x}{1+x}}$	

Another Example

Q Use the substitution $u = \sin x + 1$ to find $\int \cos x \sin x \, (1 + \sin x)^3 \; dx$

STEP 1: Using substitution, work out x and dx (or variant)

$$u = \sin x + 1$$
$$\frac{du}{dx} = \cos x \quad \rightarrow \quad du = \cos x \; dx$$
$$\sin x = u - 1$$

Notice this time we didn't find x or dx. We could, but then $x = \arcsin(u - 1)$, and it would be slightly awkward simplifying the expression (although is still very much a valid method!)

STEP 2: Substitute these into expression.

$$= \int (u - 1)u^3 \; du$$

STEP 3: Integrate simplified expression.

$$= \int u^4 - u^3 \; du$$
$$= \frac{1}{5}u^5 - \frac{1}{4}u^4 + C$$

STEP 4: Write answer in terms of x.

$$= \frac{1}{5}(\sin x + 1)^5 - \frac{1}{4}(\sin x + 1)^4 + C$$

Using substitutions involving implicit differentiation

When a root is involved, it makes thing much tidier if we use $u^2 = \cdots$

Q Use the substitution $u^2 = 2x + 5$ to find $\int x\sqrt{2x+5}\ dx$

STEP 1: Using substitution, work out x and dx (or variant)

$$2u\frac{du}{dx} = 2 \quad \rightarrow \quad dx = u\,du$$

$$x = \frac{u^2 - 5}{2}$$

STEP 2: Substitute these into expression.

$$\int x\sqrt{2x+5}\ dx = \int \frac{u^2-5}{2}u \times u du$$

$$= \int \frac{1}{2}u^4 - \frac{5}{2}u^2\ du$$

STEP 3: Integrate simplified expression.

$$= \frac{1}{10}u^5 - \frac{5}{6}u^3 + C$$

STEP 4: Write answer in terms of x.

$$= \frac{(2x+5)^{\frac{5}{2}}}{10} - \frac{5(2x+5)^{\frac{3}{2}}}{6} + C$$

This was marginally less tedious than when we used $u = 2x + 5$, as we didn't have fractional powers to deal with.

Test Your Understanding

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(c) Using the substitution $u = 1 + \cos x$, or otherwise, show that

$$\int \frac{2\sin 2x}{(1+\cos x)}\,dx = 4\ln(1+\cos x) - 4\cos x + k,$$

where k is a constant. **(5)**

Hint: You might want to use your double angle formula first.

?

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Edexcel C4 Jan 2012 Q6c

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where k is a constant. **(5)**

Hint: You might want to use your double angle formula first.

$$\frac{du}{dx} = -\sin x \qquad \rightarrow \qquad dx = -\frac{1}{\sin x}du$$

$\cos x = u - 1$ (As before $x = \arccos(u-1)$ is going to be messy)

$$\int \frac{2\sin 2x}{1+\cos x}dx = \int \frac{4\sin x\cos x}{1+\cos x}dx = \int -\frac{4\sin x\,(u-1)}{u}\frac{1}{\sin x}du$$

$$= -4\int \frac{u-1}{u}du = -4\int 1 - \frac{1}{u}du$$

$$= -4(u - \ln|u|) + c = -4(1 + \cos x - \ln|\cos x + 1|) + c = \cdots$$

Definite Integration

Now consider:

Q Calculate $\int_0^{\frac{\pi}{2}} \cos x \sqrt{1+\sin x}\ dx$

Use substitution: ?

?

Now because we've changed from x to u, we have to work out what values of u would have given those limits for x:

When $x = \frac{\pi}{2}$, $u =$?

When $x = 0$, $u =$

$$\int_0^{\frac{\pi}{2}} \cos x \sqrt{1+\sin x}\ dx = ?$$

$$=$$

Definite Integration

Now consider:

Q Calculate $\int_0^{\frac{\pi}{2}} \cos x \sqrt{1+\sin x}\ dx$

Use substitution: $\boldsymbol{u = 1 + \sin x}$

$$\frac{du}{dx} = \cos x \qquad \rightarrow \qquad du = \cos x\ dx$$
$$\sin x = u - 1$$

Now because we've changed from x to u, we have to work out what values of u would have given those limits for x:

When $x = \frac{\pi}{2}$, $\quad u = 2$
When $x = 0$, $\quad u = 1$

$$\int_0^{\frac{\pi}{2}} \cos x \sqrt{1+\sin x}\ dx = \int_1^2 u^{\frac{1}{2}}\, du$$
$$= \left[\frac{2}{3} u^{\frac{3}{2}}\right]_1^2 = \frac{2}{3}\left(2\sqrt{2} - 1\right)$$

Test Your Understanding

Edexcel C4 June 2011 Q4

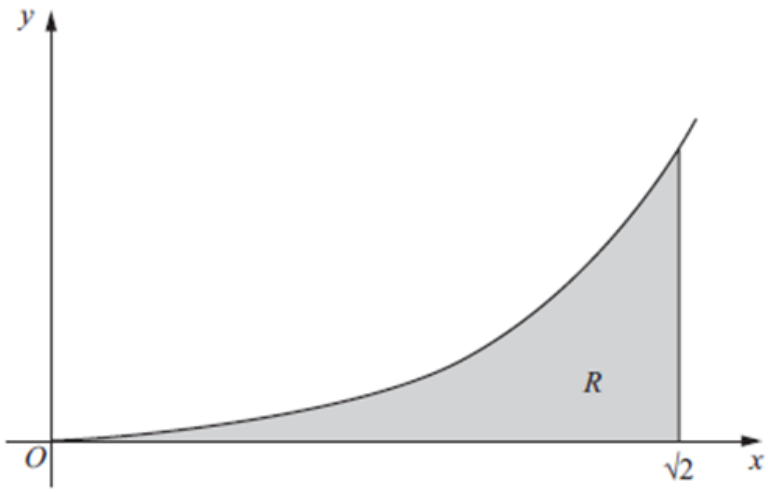

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Figure 2 shows a sketch of the curve with equation $y = x^3 \ln(x^2 + 2), x \geq 0$.

The finite region R, shown shaded in Figure 2, is bounded by the curve, the x-axis and the line $x = \sqrt{2}$.

(c) Use the substitution $u = x^2 + 2$ to show that the area of R is

$$\frac{1}{2}\int_2^4 (u-2)\ln u \, du.$$

Test Your Understanding

Edexcel C4 June 2011 Q4

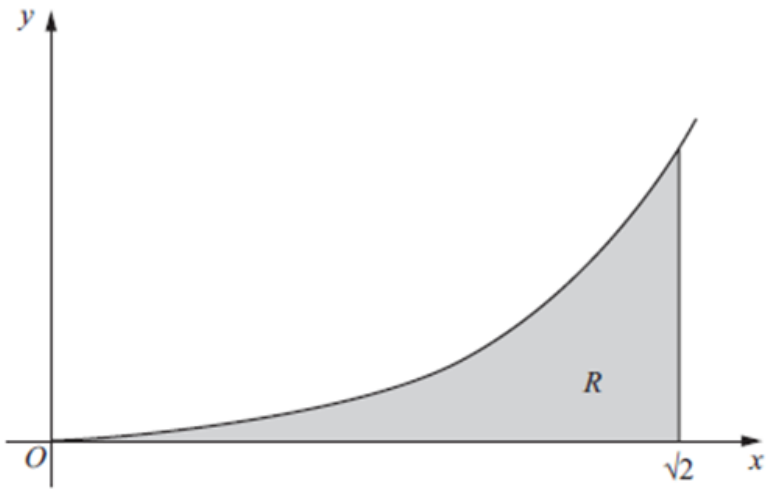

Figure 2 shows a sketch of the curve with equation $y = x^3 \ln(x^2 + 2), x \geq 0$.

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(c) Use the substitution $u = x^2 + 2$ to show that the area of R is

$$\frac{1}{2}\int_2^4 (u-2)\ln u \; du.$$

$$\frac{du}{dx} = 2x \quad \rightarrow \quad du = 2x\,dx$$

$$x^2 = u - 2$$

When $x = \sqrt{2}, u = \left(\sqrt{2}\right)^2 + 2 = 4$

When $x = 0, u = 0^2 + 2 = 2$

$$\int_0^{\sqrt{2}} x^2 \ln\left(x^2 + 2\right) x\,dx$$

$$= \int_2^4 \frac{1}{2}(u-2)\ln(u)\;du$$

Exercise 11.5

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Homework Exercise

1 Use the substitutions given to find:

a $\int x\sqrt{1+x}\,\mathrm{d}x;\ u = 1 + x$

b $\int \frac{1+\sin x}{\cos x}\,\mathrm{d}x;\ u = \sin x$

c $\int \sin^3 x\,\mathrm{d}x;\ u = \cos x$

d $\int \frac{2}{\sqrt{x}(x-4)}\,\mathrm{d}x;\ u = \sqrt{x}$

e $\int \sec^2 x \tan x\sqrt{1+\tan x}\,\mathrm{d}x;\ u^2 = 1 + \tan x$

f $\int \sec^4 x\,\mathrm{d}x;\ u = \tan x$

2 Use the substitutions given to find the exact values of:

a $\int_0^5 x\sqrt{x+4}\,\mathrm{d}x;\ u = x + 4$

b $\int_0^2 x(2+x)^3\,\mathrm{d}x;\ u = 2 + x$

c $\int_0^{\frac{\pi}{2}} \sin x\sqrt{3\cos x + 1}\,\mathrm{d}x;\ u = \cos x$

d $\int_0^{\frac{\pi}{3}} \sec x \tan x\sqrt{\sec x + 2}\,\mathrm{d}x;\ u = \sec x$

e $\int_1^4 \frac{1}{\sqrt{x}(4x-1)}\,\mathrm{d}x;\ u = \sqrt{x}$

3 By choosing a suitable substitution, find:

a $\int x(3+2x)^5\,\mathrm{d}x$

b $\int \frac{x}{\sqrt{1+x}}\,\mathrm{d}x$

c $\int \frac{\sqrt{x^2+4}}{x}\,\mathrm{d}x$

4 By choosing a suitable substitution, find the exact values of:

a $\int_2^7 x\sqrt{2+x}\,\mathrm{d}x$

b $\int_2^5 \frac{1}{1+\sqrt{x-1}}\,\mathrm{d}x$

c $\int_0^{\frac{\pi}{2}} \frac{\sin 2\theta}{1+\cos\theta}\,\mathrm{d}\theta$

5 Using the substitution $u^2 = 4x + 1$, or otherwise, find the exact value of $\int_6^{20} \frac{8x}{\sqrt{4x+1}}\,\mathrm{d}x$. **(8 marks)**

Homework Exercise

6 Use the substitution $u^2 = e^x - 2$ to show that $\int_{\ln 3}^{\ln 4} \frac{e^{4x}}{e^x - 2}\,dx = \frac{a}{b} + c\ln d$, where a, b, c and d are integers to be found. **(7 marks)**

7 Prove that $-\int \frac{1}{\sqrt{1-x^2}}\,dx = \arccos x + c$. **(5 marks)**

8 Use the substitution $u = \cos x$ to show

$\int_0^{\frac{\pi}{3}} \sin^3 x \cos^2 x\,dx = \frac{47}{480}$ **(7 marks)**

Hint Use exact trigonometric values to change the limits in x to limits in u.

9 Using a suitable trigonometric substitution for x, find $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} x^2\sqrt{1-x^2}^2\,dx$. **(8 marks)**

Challenge

By using a substitution of the form $x = k\sin u$, show that

$$\int \frac{1}{x^2\sqrt{9-x^2}}\,dx = -\frac{\sqrt{9-x^2}}{9x} + c$$

Homework Answers

1 **a** $\frac{2}{5}(1+x)^{\frac{5}{2}} - \frac{2}{3}(1+x)^{\frac{3}{2}} + c$ **b** $-\ln|1-\sin x| + c$

c $\dfrac{\cos^3 x}{3} - \cos x + c$ **d** $\ln\left|\dfrac{\sqrt{x}-2}{\sqrt{x}+2}\right| + c$

e $\frac{2}{5}(1+\tan x)^{\frac{5}{2}} - \frac{2}{3}(1+\tan x)^{\frac{3}{2}} + c$

f $\tan x + \frac{1}{3}\tan^3 x + c$

2 **a** $\frac{506}{15}$ **b** $\frac{392}{5}$ **c** $\frac{14}{9}$ **d** $\frac{16}{3} - 2\sqrt{3}$ **e** $\frac{1}{2}\ln\frac{9}{5}$

3 **a** $\dfrac{(3+2x)^7}{28} - \dfrac{(3+2x)^6}{8} + c$ **b** $\frac{2}{3}(1+x)^{\frac{3}{2}} - 2\sqrt{1+x} + c$

c $\sqrt{x^2+4} + \ln\left|\dfrac{\sqrt{x^2+4}-2}{\sqrt{x^2+4}+2}\right| + c$

4 **a** $\frac{886}{15}$ **b** $2 + 2\ln\frac{2}{3}$ **c** $2 - 2\ln 2$

5 $\frac{592}{3}$

6 $\displaystyle\int_{\ln 3}^{\ln 4} \frac{e^{4x}}{e^x - 2}\,dx = \int_1^{\sqrt{2}} \frac{2(u^2+2)^3}{u}\,du$

$\displaystyle= \int_1^{\sqrt{2}} \left(2u^5 + 12u^3 + 24u + \frac{16}{u}\right) du$

$\displaystyle= \left[\frac{1}{3}u^6 + 3u^4 + 12u^2 + 16\ln u\right]_1^{\sqrt{2}}$

$\displaystyle= \left(\frac{116}{3} + 16\ln\sqrt{2}\right) - \left(\frac{46}{3} + 16\ln 1\right)$

$\displaystyle= \frac{70}{3} + 16\ln\sqrt{2} = \frac{70}{3} + 8\ln 2 \Rightarrow$

$a = 70,\ b = 3,\ c = 8,\ d = 2$

7 $x = \cos\theta, \dfrac{dx}{d\theta} = -\sin\theta$

$\displaystyle\int -\frac{1}{\sqrt{1-x^2}}\,dx = \int -\frac{1}{\sin\theta}(-\sin\theta)\,d\theta$

$\displaystyle= \int 1\,d\theta = \theta + c = \arccos x + c$

8 $\displaystyle\int_0^{\frac{\pi}{3}} \sin^3 x\cos^2 x\,dx = \int_1^{\frac{1}{2}} (u^2-1)u^2\,du = \int_1^{\frac{1}{2}} (u^4 - u^2)\,du$

$\displaystyle= \left[\frac{1}{5}u^5 - \frac{1}{3}u^3\right]_1^{\frac{1}{2}} = \frac{47}{480}$

9 $\dfrac{2\pi + 3\sqrt{3}}{96}$

Challenge

$x = 3\sin u, \dfrac{dx}{du} = 3\cos u \Rightarrow dx = 3\cos u\,du$

$(3\sin u)^2 + (3\cos u)^2 = 9$

$\Rightarrow x^2 + (3\cos u)^2 = 9 \Rightarrow \cos u = \dfrac{\sqrt{9-x^2}}{3}$

$\displaystyle\int \frac{1}{x^2\sqrt{9-x^2}}\,dx = \int \frac{1}{9\sin^2 u\cos u}(3\cos u)\,du$

$\displaystyle= \frac{1}{9}\int \operatorname{cosec}^2 u\,du = -\frac{1}{9}\cot u + c = -\frac{\cos u}{9\sin u} + c$

$\displaystyle= -\frac{\frac{\sqrt{9-x^2}}{3}}{3x} + c = -\frac{\sqrt{9-x^2}}{9x} + c$